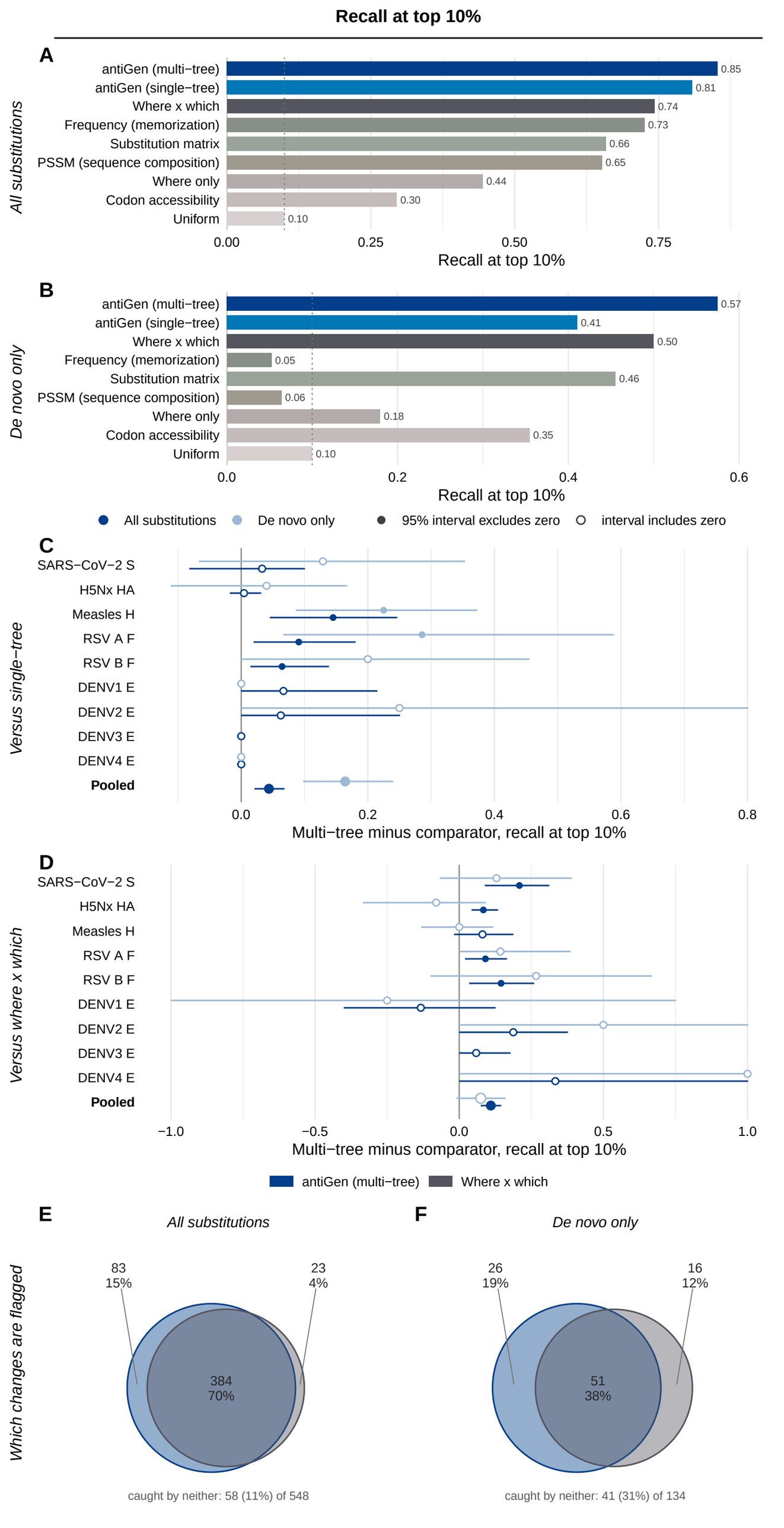
Recall at top 10%
A
All substitutions
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
Substitution matrix
PSSM (sequence composition)
Where only
Codon accessibility
Uniform
0.85
0.81
0.74
0.73
0.66
0.65
0.44
0.30
0.10
0.00
0.25
0.50
0.75
Recall at top 10%
B
De novo only
0.57
0.41
0.50
0.05
0.46
0.06
0.18
0.35
0.10
0.0
0.2
0.4
0.6
Recall at top 10%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
0.0
0.2
0.4
0.6
0.8
Multi−tree minus comparator, recall at top 10%
D
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−1.0
−0.5
0.0
0.5
1.0
Multi−tree minus comparator, recall at top 10%
antiGen (multi−tree)
Where x which
E
All substitutions
83
15%
23
4%
384
70%
caught by neither: 58 (11%) of 548
F
De novo only
26
19%
16
12%
51
38%
caught by neither: 41 (31%) of 134
Which changes are flagged